Digital Logic Design (EE1005)

Sessional-II Exam

Date: April 9th 2026

Course Instructor(s)

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Total Time (Hrs): 1
Total Marks: 45
Total Questions: 5

Roll No ______ Section ______ Student Signature ______

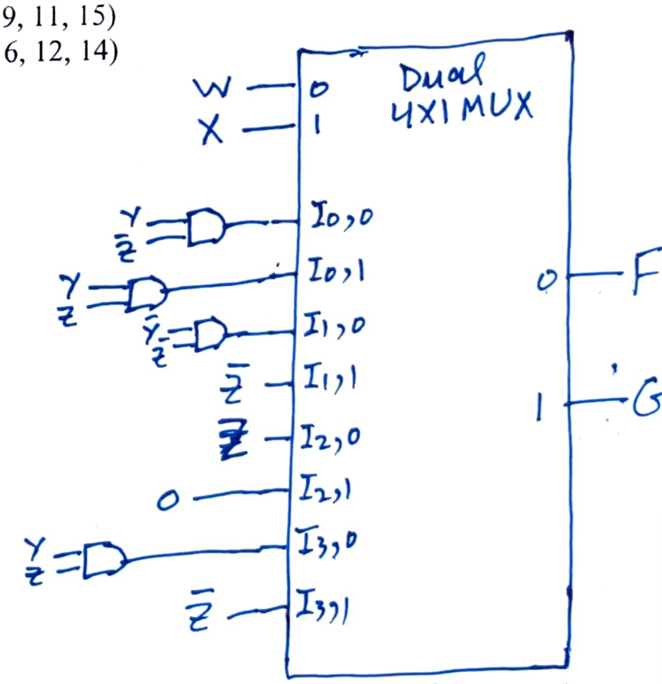
Instruction/Notes:
- *Answer all the questions in the space provided.* ***NO answer booklet is required.***
- *Be precise and the answer to each question should be readable.*
- *You can use* ***rough sheet*** *but* ***DO NOT ATTACH IT***.
- ***Use of calculator is not allowed.***

CLO # 2: **Construct optimized combinational circuit design.**

Question#1 (a): Implement following function using **a dual 4x1 multiplexer** and minimum external gates taking W and X as selection input. Properly label your circuit to get credit. [7+5]

$$F(W,X,Y,Z) = \sum m(2, 4, 9, 11, 15)$$
$$G(W,X,Y,Z) = \sum m(3, 4, 6, 12, 14)$$

WXYZ	F		G	
0000	0		0	
0001	0	$Y\bar{Z}$	0	YZ
0010	1		0	
0011	0		1	
0100	1		1	
0101	0	$\bar{Y}\bar{Z}$	0	$\bar{Z}$
0110	0		1	
0111	0		0	
1000	0		0	
1001	1	Z	0	0
1010	0		0	
1011	1		0	
1100	0		1	
1101	0	YZ	0	$\bar{Z}$
1110	0		1	
1111	1		0	



b) You are designing a digital control system for a hospital emergency room. There are four patient call buttons, each representing a different level of medical urgency. Since medical staff can attend to only one patient at a time, the system must output the **binary code of the highest-priority active input** whenever one or more buttons are pressed simultaneously.

The Priority Levels:

The buttons are ranked from highest priority (3) to lowest priority (0):

1. Level 3 (Critical): Immediate life-threatening condition.
2. Level 2 (Urgent): Serious condition requiring quick intervention.
3. Level 1 (Stable): Non-life-threatening, but requires attention.
4. Level 0 (Routine): Minor check-up or inquiry.

Please provide truth table for the required digital control system only.

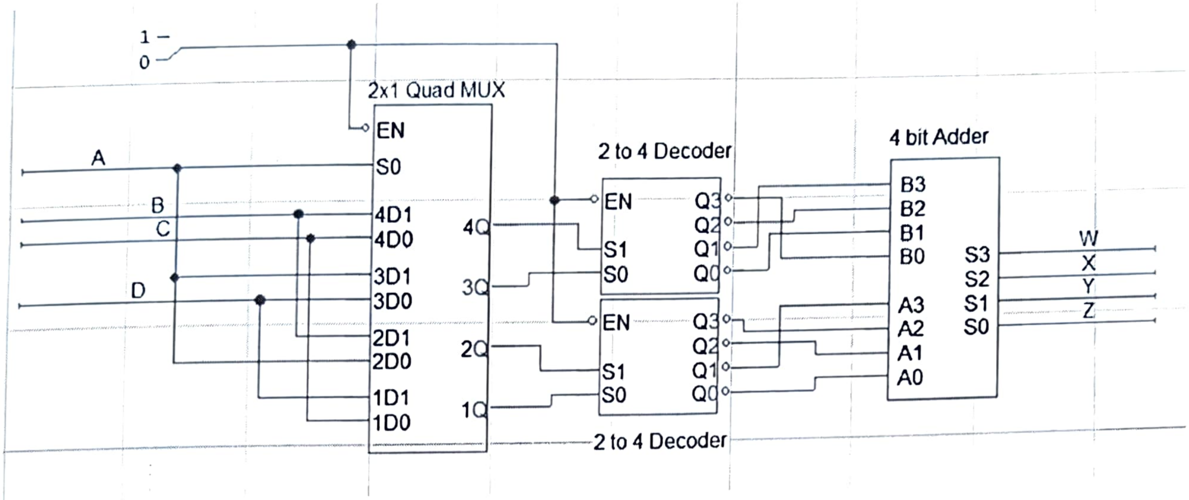
Priority levels				Binary code		Validity
L3	L2	L1	L0	A1	A0	V
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	1
0	0	1	1	0	1	1
0	1	0	0	1	0	1
0	1	0	1	1	0	1
0	1	1	0	1	0	1
0	1	1	1	1	0	1
1	0	0	0	1	1	1
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

Compressed Table

L3	L2	L1	L0	A1	A0	V
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

CLO # 2: Construct optimized combinational circuit diagram.

Question#2 (a): Fill in the Truth Table according to the following circuit diagram. **[15]**



A	B	C	D	1Q	2Q	3Q	4Q	B3	B2	B1	B0	A3	A2	A1	A0	W	X	Y	Z
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	1	1
0	0	0	1	0	0	1	0	1	0	0	0	0	0	0	1	1	0	0	1
0	0	1	0	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0	0
0	0	1	1	1	0	1	1	0	0	0	1	1	0	0	0	1	0	0	1
0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	1	1
0	1	0	1	0	0	1	0	1	0	0	0	0	0	0	1	1	0	0	1
0	1	1	0	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0	0
0	1	1	1	1	0	1	1	0	0	0	1	1	0	0	0	1	0	0	1
1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	1	1	0	0	1
1	0	0	1	1	0	1	0	1	0	0	0	1	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	1	0	0	0	0	0	0	1	1	0	0	1
1	0	1	1	1	0	1	0	1	0	0	0	1	0	0	0	0	0	0	0
1	1	0	0	0	1	1	1	0	0	0	1	0	0	1	0	0	0	1	1
1	1	0	1	1	1	1	1	0	0	0	1	0	1	0	0	0	1	0	1
1	1	1	0	0	1	1	1	0	0	0	1	0	0	1	0	0	0	1	1
1	1	1	1	1	1	1	1	0	0	0	1	0	1	0	0	0	1	0	1

CLO#2: Construct optimized combinational circuit diagram.

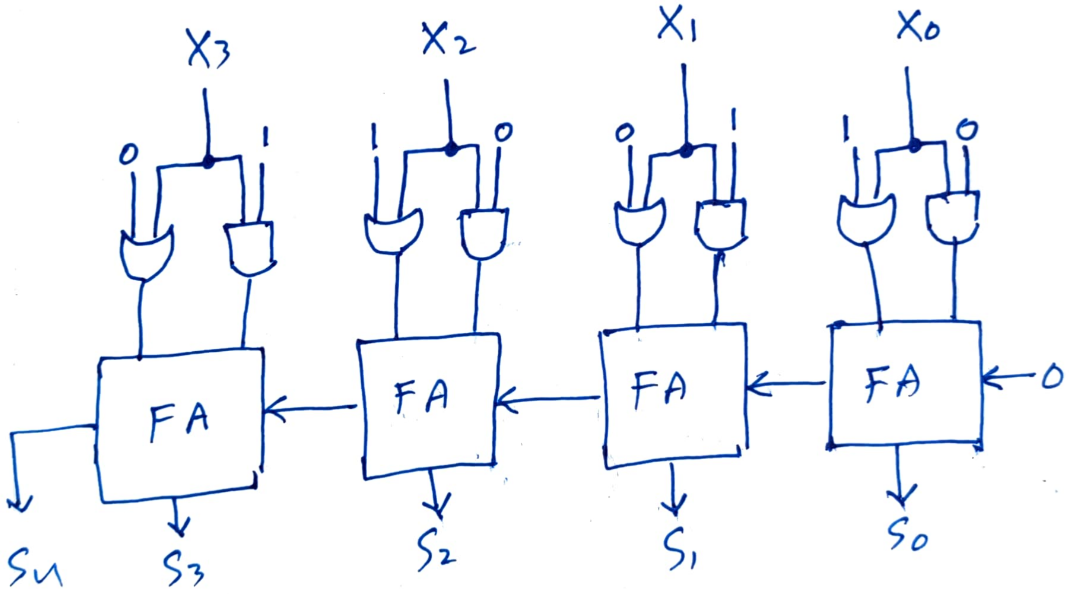
Question#3 (a): Design a combinational circuit using an iterative design approach to compute the following for a 4-bit binary input $X=(X_3,X_2,X_1,X_0)$. **[9+9]**

- Let A=(0101), a fixed 4-bit constant.
- Let B=(1010), another fixed 4-bit constant.
- The circuit should first perform a **bitwise OR** operation between X and A.
- Simultaneously, it should perform a **bitwise AND** operation between X and B.
- Finally, the circuit should **add the two results together** using iterative full-adder stages.

Requirements:

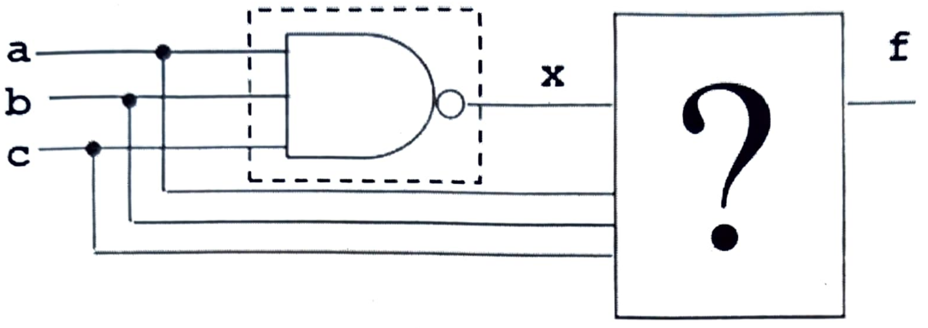
1. The circuit must be designed without using a truth table.
2. Use repetitive (iterative) units to simplify the hardware implementation.
3. Clearly define the functionality of a single stage unit (for one bit).
4. Connect multiple stages to form the complete 4-bit circuit.

Draw a block diagram showing inputs, outputs, and intermediate connections.



b) Design a circuit that verifies the logical operation of a 3-input NAND gate. f = '1' (LED ON) if the NAND gate does NOT work properly. You must put some combinations of gates in place of question mark in the figure. ***Assumption:*** when the NAND gate is not working, it generates 1's instead of 0's and vice versa.

Draw Truth table after identifying the inputs and outputs. Optimize the equation and draw logic diagram using the gates that will replace question mark in the diagram given below.



a b c x	f
0 0 0 0	1
0 0 0 1	0
0 0 1 0	1
0 0 1 1	0
0 1 0 0	1
0 1 0 1	0
0 1 1 0	1
0 1 1 1	0
1 0 0 0	1
1 0 0 1	0
1 0 1 0	1
1 0 1 1	0
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	1

f

ab \ cx	00	01	11	10
00	1 (0)	(1)	(3)	1 (2)
01	1 (4)	(5)	(7)	1 (6)
11	1 (12)	(13)	1 (15)	(14)
10	1 (8)	(9)	(11)	1 (10)

$$f = \bar{c}\bar{x} + \bar{a}\bar{x} + \bar{b}\bar{x} + abcx$$
$$= \bar{x}(\bar{a} + \bar{b} + \bar{c}) + x\,abc$$
$$= \bar{x}(\overline{abc}) + x(abc)$$
$$= \overline{x \oplus abc}$$